Web Wii

A Modern Wii Remote Recreation
Using IR Tracking and BLE

Mika Baëza Mohammad Parto Kawtar Bidoukhach

SEES Project – 2026

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Why did we build this project?

- Inspired by the Nintendo Wii and its intuitive motion-based interaction
- Recreate a modern Wii-like pointing experience
- Use embedded hardware for real-time motion tracking
- Enable wireless interaction directly through a web browser

Goal: Create a fully wireless motion-controlled pointing system.

Background: The Original Wii Remote

- Wii Remote uses a **passive IR sensor bar** + on-board camera to track cursor position
- Original motion sensing: 3-axis **accelerometer** — no rotational tracking
- We keep the same IR architecture but replace the sensor bar with **two fixed IR LEDs**
- Instead of raw accelerometer data, we use a **BNO085 IMU** — outputs Euler angles directly
- Orientation is ready to use, no manual sensor fusion needed

Result: Simpler firmware, more reliable cursor tracking.

Challenges and Solutions

Challenge 1 — Depth Ambiguity

When using a single IR point, the controller distance cannot be estimated reliably.

▶ **Solution:** Use two IR LEDs with fixed spacing to recover depth through geometry.

Challenge 2 — Orientation Stability

Raw gyroscope measurements accumulate drift over time.

▶ **Solution:** Use the BNO085 onboard sensor fusion and reset calibration.

Challenge 3 — Real-Time Communication

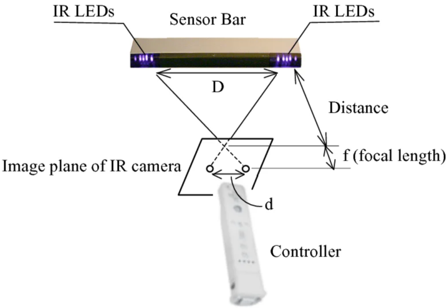
Cursor updates must remain responsive and low latency.

▶ **Solution:** Use BLE notifications at approximately 100 Hz.

System Architecture

- Two IR LEDs are placed near the screen
- The remote camera detects the LEDs
- The IMU measures controller orientation
- The ESP32 computes cursor position
- BLE sends cursor data to the webpage

IR LEDs → Camera → ESP32

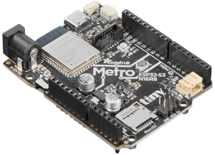


Remote Components

- ESP32-S3
- BNO085 IMU (UART protocol)
- IR Camera (I2C protocol)
- Push Buttons
- Breadboard Prototype

Sensor Bar

- 2 IR LEDs
- Fixed spacing



ESP32-S3

BNO085 IMU

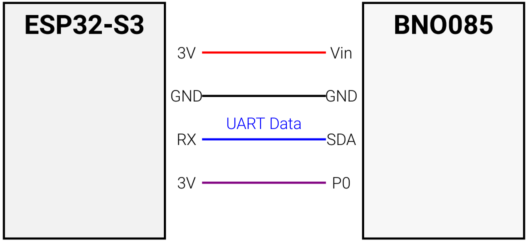
IR Camera

IR LED

BNO085 Wiring to ESP32-S3 (UART-RVC)

Why UART-RVC instead of I2C?

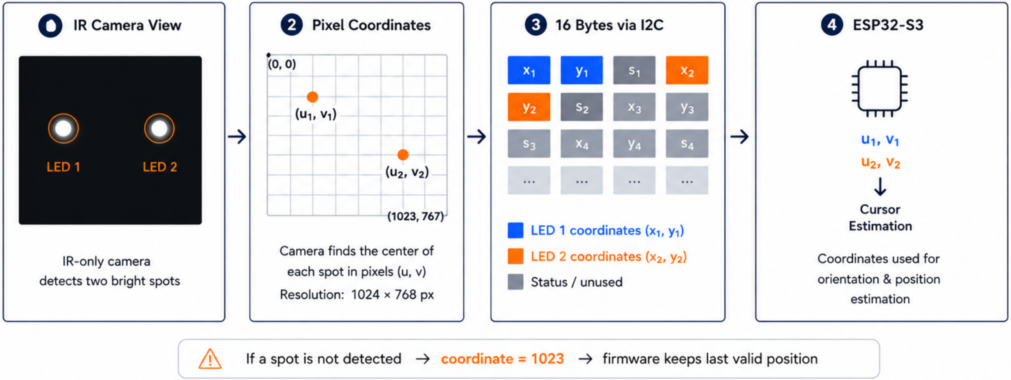
- BNO085 I2C is unreliable on ESP32-S3
- UART-RVC streams orientation continuously
- Only one wire is needed for data
- No sensor configuration required



P0 HIGH enables UART-RVC mode

Baudrate: `115200`

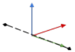
How the IR Camera Works



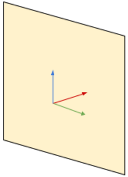
Computation overview

Definitions

World frame:
Reference frame in which the computation are made.



Reference leds :
Separated of 20cm and centered in the world frame.



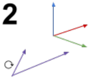
Screen plane:
plane of coordinate Y=0

Steps

1

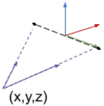
What camera sees.

2



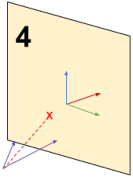
Reconstruct vectors in 3D space, from known camera FOV and IMU rotation.

3



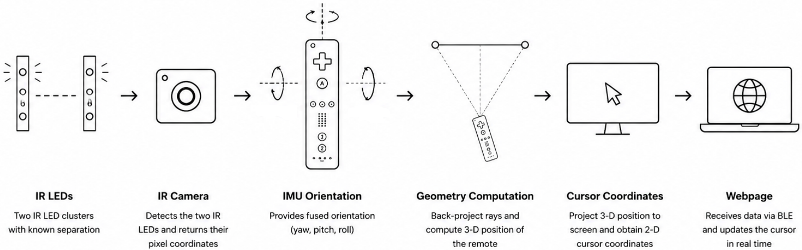
Compute position thanks to known led spacing

4



From known position and rotation, calculate the cursor position

Cursor Localization Pipeline



Computation overview

The remote transmits a 14-byte packet to the console at ~100 Hz:

```
struct RemotePacket {
    int32_t x;              // mm * 100          (4 bytes)
    int32_t y;              // mm * 100          (4 bytes)
    int32_t pitch;          // degrees * 100     (4 bytes)
    uint8_t buttonMask;     // bitmask           (1 byte)
    uint8_t missMask;       // bitmask           (1 byte)
};                          // Total: 14 bytes
```

Listing 1: BLE packet structure (remote → PC)

The `MISS_MASK`

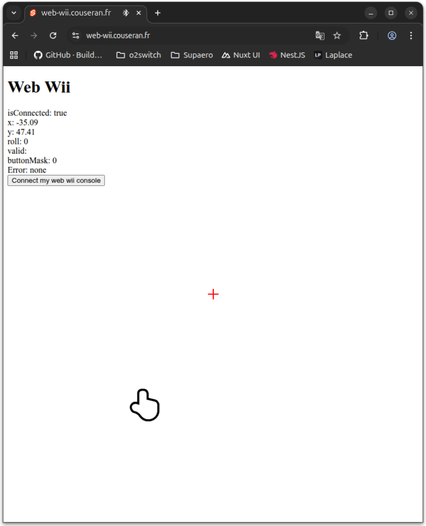
Bit 0	**Bit 1**	**Bit 2**
IMU Miss	IR Camera Miss	Localization Fail
BNO085 read failed	IR LEDs not detected	Invalid 3D estimation

`missMask = 0` $\rightarrow$ valid packet

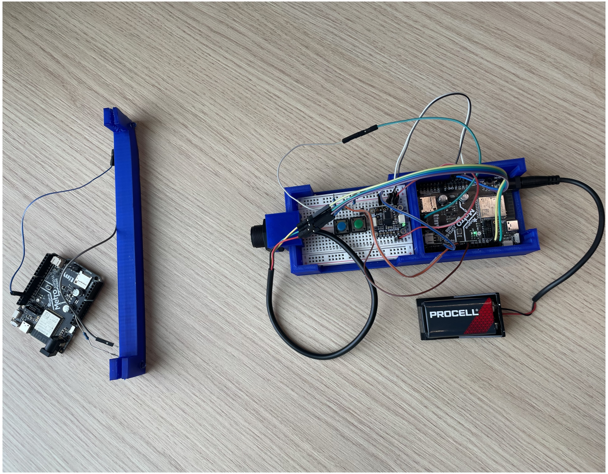
`missMask` $\neq$ `0` $\rightarrow$ packet ignored

Web Interface

- A browser-based interface was developed using Svelte and SvelteKit
- The webpage communicates directly with the remote using the Web Bluetooth API
- Cursor coordinates and button states are updated in real time through BLE communication
- No native desktop application or driver is required
- Compatible with Chromium-based browsers such as Chrome and Edge



Final Prototype



Limitations and Future Work

- **IR LED emission angle:** The current IR LEDs have a limited emission cone, causing tracking loss when the controller is pointed too far away from the screen axis. Wider-angle IR LEDs could significantly improve robustness.
- **BNO085 sensor fusion latency:** The onboard fusion algorithm prioritizes orientation stability, introducing a small delay during fast rotational movements.
- **Browser compatibility:** The system relies on the Web Bluetooth API, which is currently supported mainly by Chromium-based browsers such as Chrome and Edge.
- **Future improvements:** Potential improvements include wider-angle LEDs, PCB integration for better hardware stability, and software filtering for smoother cursor tracking.

Conclusion

This project successfully recreated the core Nintendo Wii pointing experience using modern embedded hardware, IR tracking, IMU orientation sensing, BLE communication, and browser-based rendering. The final system achieved real-time cursor interaction without requiring native desktop software, while demonstrating that a modern motion-controlled interface can be built using accessible low-cost components and web technologies.

Thank You!

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